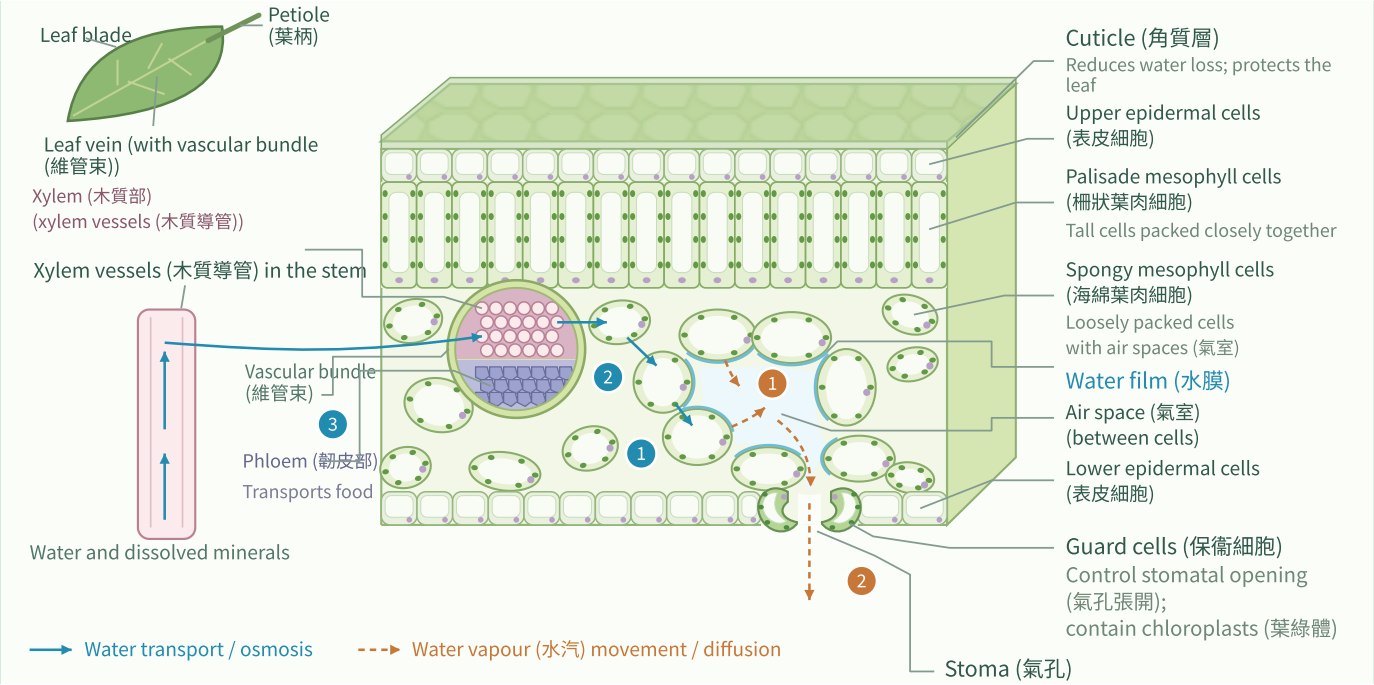
Name: _____ Class and student number: _____

Red water traces and water transport in celery

Read A from the start and while doing B. Receive C only after completing and keeping answers in B.

Read first: water transport and transpiration (蒸騰) in a leaf



Transpiration (蒸騰) through stomata (氣孔)

1 Water evaporates (蒸發) from the surfaces of mesophyll cells (葉肉細胞) into the air spaces (氣室).

2 Water vapour (水汽) in the air spaces (氣室) diffuses through stomata (氣孔) into the air.

How transpiration pull (蒸騰拉力) forms

1 Mesophyll cells (葉肉細胞) lose water, so their water potential falls.

2 Mesophyll cells (葉肉細胞) take in water from nearby cells by osmosis.

3 Water is drawn out of the xylem vessels (木質導管), creating transpiration pull (蒸騰拉力).

1. Transpiration (蒸騰) is the process in which water in a plant evaporates (蒸發) and is lost to the air as water vapour (水汽). Water vapour (水汽) mainly leaves through stomata (氣孔) in leaves and herbaceous stems (草本莖).
   a. Water evaporates (蒸發) from the water film (水膜) on the surface of mesophyll cells (葉肉細胞) and enters the air spaces (氣室) in the spongy mesophyll as water vapour (水汽). When the water vapour (水汽) concentration in the air spaces (氣室) is higher than in the air outside, water vapour (水汽) leaves the leaf through stomata (氣孔) by diffusion.
   b. When mesophyll cells (葉肉細胞) lose water, their water potential falls. They take in water from nearby cells by osmosis. This draws water out of the leaf's xylem vessels (木質導管) and creates transpiration pull (蒸騰拉力), which pulls water upwards through the stem's xylem vessels (木質導管).
2. Under suitable conditions, such as enough water being available, higher light intensity (光強度) usually makes stomata (氣孔) open wider. Water vapour (水汽) then leaves the leaf faster by diffusion, so the transpiration rate (蒸騰速率) increases.
   Note: More light may also make the leaf's temperature rise, increasing transpiration (蒸騰) further. This investigation therefore assumes that all setups have the same temperature so we can compare the effect of light intensity (光強度).
3. Transpiration pull (蒸騰拉力) keeps roots taking in water and moves water and dissolved minerals absorbed by the roots upwards through xylem vessels (木質導管) to the leaves.

See textbook section 11.1, pages 11-3 to 11-8. The red water trace gives evidence of water transport. Its upward speed is not a direct measure of the transpiration rate (蒸騰速率). Comparing celery with and without leaves alone also cannot prove the role of stomata (氣孔).

Internal leaf structure and transpiration (蒸騰)

Orange numbers: transpiration (蒸騰) through stomata (氣孔) | Blue numbers: how transpiration pull (蒸騰拉力) forms

Enlarged view, not to scale. The celery stalk in this experiment is a petiole (葉柄). The stem shown here explains transport through xylem vessels (木質導管).

Worked methods (not answers for A–D)

Explanation → worked example → supported practice → independent application → checking and correction

Distinguish processes and plant parts

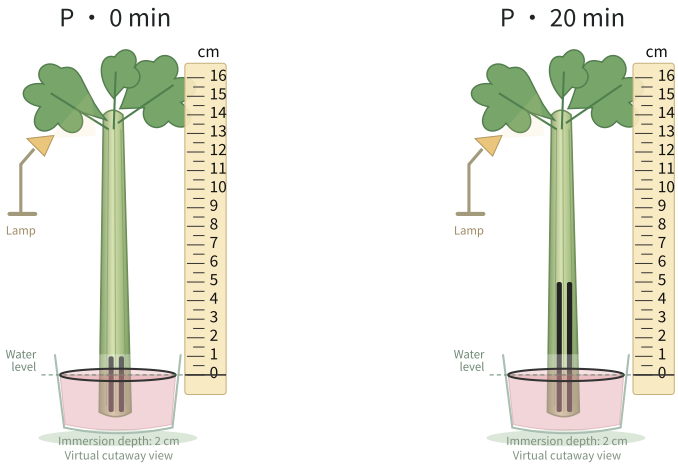
Water transport is liquid water moving through xylem (木質部). Transpiration (蒸騰) is water loss from a plant to the air. The trace gives evidence of liquid water transport, not a direct measurement of water vapour (水汽) lost. The long celery stalk is a petiole (葉柄) containing xylem (木質部), not a stem.

Hypothesis and fair-comparison example

A testable (可測試的) hypothesis states the factor changed on purpose and a measurable result. Example: “With other conditions the same, celery with a larger leaf area has a faster average upward speed of the red trace.” Compare two leaf areas, keep light and other conditions the same, and measure in cm/min. Results may not support the hypothesis.

Independent variable: the factor changed on purpose. Dependent variable: the measured result. Controlled variables: factors kept the same. A fair comparison changes only the factor being studied. When comparing leaf area, keep light, temperature and measurement time the same.

Ruler and calculation example



Neutral practice sample P (a separate method example): the left image is 1.0 cm at 0 min; the right image is 5.0 cm at 20 min. Read the ruler at the level of the trace top. The liquid surface is the fixed zero; each small division is 0.5 cm.

Example average speed = (5.0 − 1.0) ÷ 20 = 0.20 cm/min. Do not simply divide final height by time; subtract the starting reading first.

Plotting example

Another method example: light intensity (光強度) 500 lux, speed 0.20 cm/min. Find 500 on X and 0.20 on Y; mark their intersection. These values teach the method only, not investigation setup data.